

Dual Pipeline Comparison Report

152,909 trips in unseen test-quarter set were compared, way more than the required minimum of 100, recorded into dual_pipeline_comparison.csv file, one record per trip: trip id, ground truth, Spark prediction, Python prediction, agreement/non-agreement, probability of prediction, numerical probability difference, consistency indicator, and the reasoning for inconsistency.

Metric	Value
Records compared	152,909
Overall agreement	94.1%
Probability correlation	0.9378
Disagreements	9,028
of which, borderline (both near threshold)	5,606
of which, materially inconsistent	3,422
Spark correct when they disagree	2,888
Python correct when they disagree	6,140
Mean probability difference	-0.014

Independently trained models are expected not to be alike, 94.1% similarity and 0.94 probability correlation indicate that both models have learned the same algorithmic pattern. In cases where they do not agree on a given data point, the Python model (gradient-boosted) is right more frequently, which reflects its superiority among the chosen algorithms.